MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 03
Chapter 11: Areas Related to Circles

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE

MATHS MANTHAN | (5 × 1 = 5 Marks)

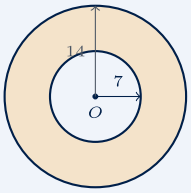
Q1. The area enclosed between two concentric circles of radii 7 cm and 14 cm is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 462 cm^2 (B) 308 cm^2 (C) 154 cm^2 (D) 616 cm^2

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Formula for Annular Ring [0.5 Mark]

Let $R = 14$ cm and $r = 7$ cm.



$$\text{Area} = \pi(R^2 - r^2) = \pi(R - r)(R + r)$$

Step 2: Factoring and Calculation [0.5 Mark]

$$\text{Area} = \frac{22}{7} \times (14 - 7)(14 + 7) = \frac{22}{7} \times 7 \times 21 = 22 \times 21 = 462 \text{ cm}^2$$

Correct Option: (A) 462 cm^2

Enclosed Area = **462** cm^2

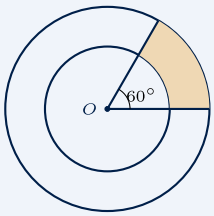
Q2. There are two concentric circles of radii 8 cm and 5 cm. A sector forms an angle of 60° at the common centre O. The area of the shaded sector annulus is: [1]

(A) $\frac{77}{2}\pi$ cm^2 (B) $\frac{13}{2}\pi$ cm^2 (C) $\frac{39}{4}\pi$ cm^2 (D) $\frac{65}{6}\pi$ cm^2

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Step 1: Formula for Sector of Annulus [0.5 Mark]

Outer radius $R = 8$ cm, inner radius $r = 5$ cm, central angle $\theta = 60°$.



$$\text{Area} = \frac{\theta}{360°} \times \pi(R^2 - r^2)$$

Step 2: Substitution and Simplification [0.5 Mark]

$$\text{Area} = \frac{60°}{360°} \times \pi(8^2 - 5^2) = \frac{1}{6} \times \pi \times (64 - 25) = \frac{39}{6}\pi = \frac{13}{2}\pi \text{ cm}^2$$

Correct Option: (B) $\frac{13}{2}\pi$ cm²

$$\boxed{\text{Area of Sector Annulus} = \mathbf{\frac{13}{2}\pi \text{ cm}^2}}$$

Q3. A wheel of diameter 42 cm makes 1000 revolutions. The total distance covered by the wheel is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 1.32 km (B) 13.2 km (C) 132 m (D) 1320 cm

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Step 1: Distance Covered in One Revolution [0.5 Mark]

Diameter $d = 42$ cm $\implies$ $r = 21$ cm.

$$\text{Circumference} = \pi d = \frac{22}{7} \times 42 = 22 \times 6 = 132 \text{ cm} = 1.32 \text{ m}$$

Step 2: Distance in 1000 Revolutions and Unit Conversion [0.5 Mark]

$$\text{Total Distance} = 1000 \times 1.32 \text{ m} = 1320 \text{ m} = \frac{1320}{1000} \text{ km} = 1.32 \text{ km}$$

Correct Option: (A) 1.32 km

$$\boxed{\text{Total Distance} = \mathbf{1.32} \text{ km}}$$

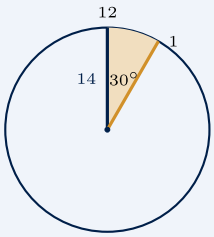
Q4. The minute hand of a clock is 14 cm long. The area swept by the minute hand in 5 minutes is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 51.33 cm² (B) 154 cm² (C) 25.67 cm² (D) 77 cm²

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Step 1: Angle Swept in 5 Minutes [0.5 Mark]

In 60 minutes, the minute hand rotates through 360°:



$$\theta = \frac{360°}{60} \times 5 = 6° \times 5 = 30°$$

Step 2: Area of the Sector [0.5 Mark]

Radius $r = 14$ cm.

$$\text{Area} = \frac{30°}{360°} \times \frac{22}{7} \times 14 \times 14 = \frac{1}{12} \times 22 \times 28 = \frac{616}{12} = \frac{154}{3} \approx 51.33 \text{ cm}^2$$

Correct Option: (A) 51.33 cm²

$$\boxed{\text{Area swept} = \mathbf{51.33} \text{ cm}^2 \quad \left(\text{or } \mathbf{\frac{154}{3}} \text{ cm}^2\right)}$$

Q5. **Assertion (A):** The total perimeter of a semicircular protractor of radius r is $(\pi + 2)r$.
Reason (R): The perimeter of a semicircular region consists of the length of the semicircular arc plus the length of its bounding diameter. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Evaluation of Reason (R) [0.5 Mark]

A closed semicircular boundary comprises the curved semicircular arc and the straight diameter line. Thus Reason (R) is **True**.

Step 2: Evaluation of Assertion (A) [0.5 Mark]

$$\text{Perimeter} = \text{Arc length} + \text{Diameter} = \pi r + 2r = (\pi + 2)r$$

Thus Assertion (A) is **True**, and Reason (R) is the exact justification.

Correct Option: **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

Correct Option: (**A**)

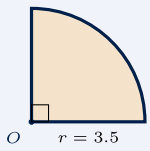
SECTION B: Very Short Answer Type

MATHS MANTHAN | **(1 × 2 = 2 Marks)**

Q6. Find the area of a quadrant of a circle whose circumference is 22 cm. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Determine Radius from Circumference [1 Mark]



$$\text{Circumference} = 2\pi r = 22 \text{ cm}$$

$$2 \times \frac{22}{7} \times r = 22 \implies r = \frac{22 \times 7}{2 \times 22} = \frac{7}{2} = 3.5 \text{ cm}$$

Step 2: Calculate Area of Quadrant [1 Mark]

$$\text{Area of quadrant} = \frac{1}{4}\pi r^2 = \frac{1}{4} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = \frac{11 \times 7}{8} = \frac{77}{8} = 9.625 \text{ cm}^2$$

Area of the quadrant = **9.625** cm^2 $\left(\text{or } \frac{\mathbf{77}}{\mathbf{8}} \text{ cm}^2\right)$

SECTION C: Short Answer Type

MATHS MANTHAN | **(3 × 3 = 9 Marks)**

Q7. A brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle into 10 equal sectors. Find:

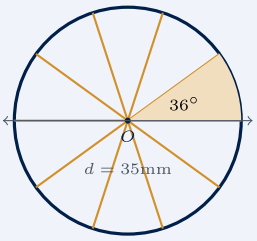
(i) the total length of the silver wire required, and

(ii) the area of each sector of the brooch. $\left(\text{Use } \pi = \frac{22}{7}\right)$

[3]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given: Diameter $d = 35$ mm $\implies$ $r = \frac{35}{2}$ mm. Number of diameters $= 5$.



Part (i): Total Length of Silver Wire [1.5 Marks]

The wire forms the circular boundary plus 5 diameters:

$$\text{Length} = \text{Circumference} + 5 \times \text{Diameter} = \pi d + 5d$$

$$\text{Length} = \left(\frac{22}{7} \times 35\right) + (5 \times 35) = (22 \times 5) + 175 = 110 + 175 = 285 \text{ mm}$$

Part (ii): Area of Each Sector [1.5 Marks]

The 5 diameters divide the circle into 10 equal sectors:

$$\text{Area of each sector} = \frac{1}{10} \times \text{Total Area} = \frac{1}{10} \times \pi r^2$$

$$= \frac{1}{10} \times \frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} = \frac{1}{10} \times 11 \times 5 \times \frac{35}{2} = \frac{385}{4} = 96.25 \text{ mm}^2$$

(i) Wire Length = **285** mm, (ii) Area of Each Sector = **96.25** mm^2 $\left(\text{or } \mathbf{\frac{385}{4}} \text{ mm}^2\right)$

Q8. A horse is tied to a peg at one corner of a square grass field of side 15 m by a 5 m long rope. Find:

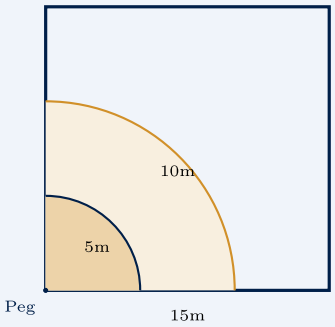
(i) the area of that part of the field in which the horse can graze, and

(ii) the increase in the grazing area if the rope were 10 m long instead of 5 m. (Use $\pi = 3.14$)

[3]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

The peg is fixed at the corner of a square grass field ($\angle = 90°$, a quadrant).



Part (i): Grazing Area with $r_1 = 5$ m Rope [1.5 Marks]

$$\text{Area}_1 = \frac{90°}{360°} \times \pi r_1^2 = \frac{1}{4} \times 3.14 \times (5)^2 = \frac{1}{4} \times 78.5 = 19.625 \text{ m}^2$$

Part (ii): Increase in Grazing Area with $r_2 = 10$ m Rope [1.5 Marks]

$$\text{Area}_2 = \frac{1}{4} \times \pi r_2^2 = \frac{1}{4} \times 3.14 \times (10)^2 = \frac{314}{4} = 78.5 \text{ m}^2$$

$$\text{Increase in Area} = \text{Area}_2 - \text{Area}_1 = 78.5 - 19.625 = 58.875 \text{ m}^2$$

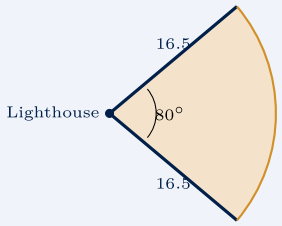
(i) Initial Area = **19.625** m^2, (ii) Increase in Grazing Area = **58.875** m^2

Q9. To warn ships for underwater rocks, a coastal lighthouse spreads a red colored light over a sector of angle $80°$ to a distance of 16.5 km. Find the area of the sea over which the ships are warned. (Use $\pi = 3.14$) **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Central angle $\theta = 80°$, Radius $r = 16.5$ km.

Step 1: Formula Declaration [1 Mark]



$$\text{Area of sector} = \frac{\theta}{360°} \times \pi r^2$$

Step 2: Substitution and Simplification [1.5 Marks]

$$\text{Area} = \frac{80°}{360°} \times 3.14 \times (16.5)^2 = \frac{2}{9} \times 3.14 \times 272.25$$

$$\text{Area} = \frac{2 \times 854.865}{9} = \frac{1709.73}{9} = 189.97 \text{ km}^2$$

Step 3: Boxed Answer with Units [0.5 Mark]

$$\boxed{\text{Area of the sea warned} = \mathbf{189.97} \text{ km}^2}$$

SECTION D: Case-Based / Applied Situation MATHS MANTHAN | (1 × 4 = 4 Marks)

Q10. Case Study: Olympic Archery Target Concentric Scoring Bands

An archery target has five concentric scoring zones from the centre outward: Gold, Red, Blue, Black, and White. The diameter of the central Gold zone is 21 cm, and each of the outer four rings has a uniform radial width of 10.5 cm.

(i) Determine the radius of the outer boundary of the Red scoring ring. **[1]**

(ii) Calculate the area of the central Gold scoring region. **[1]**

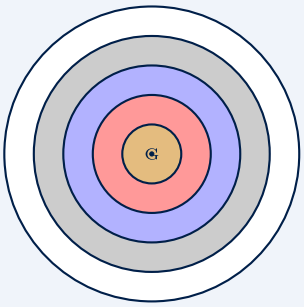
(iii) Calculate the surface area of the Red scoring band. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[2]**

— OR —

Show that the area of the Red band is exactly three times the area of the Gold region. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Given: Diameter of Gold zone $d_1 = 21$ cm $\implies$ $r_1 = 10.5$ cm. Radial width $w = 10.5$ cm.



Part (i): Radius of Outer Boundary of Red Ring [1 Mark]

$$r_2 = r_1 + w = 10.5 + 10.5 = 21 \text{ cm}$$

Part (ii): Area of Central Gold Region [1 Mark]

$$\text{Area}_{\text{Gold}} = \pi r_1^2 = \frac{22}{7} \times (10.5)^2 = \frac{22}{7} \times 110.25 = 22 \times 15.75 = 346.5 \text{ cm}^2$$

Part (iii): Area of Red Scoring Band [2 Marks]

$$\text{Area}_{\text{Red}} = \pi(r_2^2 - r_1^2) = \frac{22}{7} \times (21^2 - 10.5^2) = \frac{22}{7} \times (441 - 110.25)$$

$$= \frac{22}{7} \times 330.75 = 22 \times 47.25 = 1039.5 \text{ cm}^2$$

— OR (Alternative Choice) —

Proof that $\text{Area}_{\text{Red}} = 3 \times \text{Area}_{\text{Gold}}$: [2 Marks]

Since $r_2 = 21 = 2 \times 10.5 = 2r_1$:

$$\text{Area}_{\text{Red}} = \pi(r_2^2 - r_1^2) = \pi((2r_1)^2 - r_1^2) = \pi(4r_1^2 - r_1^2) = 3\pi r_1^2$$

$$\text{Area}_{\text{Red}} = 3 \times \text{Area}_{\text{Gold}} = 3 \times 346.5 = 1039.5 \text{ cm}^2$$

Hence proved.

(i) $r_2 = \mathbf{21}$ cm, (ii) $\text{Area}_{\text{Gold}} = \mathbf{346.5}$ cm^2, (iii) $\text{Area}_{\text{Red}} = \mathbf{1039.5}$ cm^2 $(= \mathbf{3} \times \text{Area}_{\text{Gold}})$

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